

CS549.TT Midterm #2

I. QUESTION (15 POINTS)

Given the following documents:

Doc1: ali topu at
 Doc2: ali ayşeye çiçek al
 Doc3: ali topu sev
 Doc4: ayşe topu at
 Doc5: ali ayşeye topu at

- Calculate the term-document matrix.
- Calculate the term-term matrix for a word window of ± 1 .
- Using the term-term matrix in b., calculate the cosine similarity between words ali and ayşe.

II. QUESTION (15 POINTS)

Given the following grammar L_1 below:

$S \rightarrow NP VP$
 $NP \rightarrow Pronoun$
 $NP \rightarrow ProperNoun$
 $NP \rightarrow Det Nominal$
 $Nominal \rightarrow Nominal Noun$
 $Nominal \rightarrow Noun$
 $VP \rightarrow Verb$
 $VP \rightarrow Verb NP$
 $VP \rightarrow Verb NP PP$
 $VP \rightarrow Verb PP$
 $VP \rightarrow VP PP$
 $PP \rightarrow Prep NP$
 $Noun \rightarrow \dots$
 $Verb \rightarrow \dots$
 $Pronoun \rightarrow \dots$
 $ProperNoun \rightarrow \dots$
 $Det \rightarrow \dots$
 $Prep \rightarrow \dots$

- Fill the spaces with appropriate entries to create a grammar for which a parse tree for "Alaska is a morning trip" can be constructed.
- Construct the parse tree for that parse tree in a.

III. QUESTION (15 POINTS)

Give Turkish and English examples for the following WordNet relations: Hypernym, Instance Hypernym, Part Meronym, Antonym, Derivation.

IV. QUESTION (21 POINTS)

Given the following grammar L_2 in CNF form:

$$\begin{aligned}
 S &\rightarrow NP VP \\
 NP &\rightarrow Det Nominal \\
 Nominal &\rightarrow Noun Noun \\
 VP &\rightarrow Verb NP \\
 VP &\rightarrow Verb PP \\
 VP &\rightarrow VP PP \\
 PP &\rightarrow Prep NP \\
 Noun &\rightarrow flight, trip, super, morning \\
 Verb &\rightarrow is \\
 Det &\rightarrow the, this \\
 Prep &\rightarrow on
 \end{aligned}$$

Parse the sentence "The morning flight is on this super trip" using CKY parsing. Show your work.

V. QUESTION (18 POINTS)

Given the following sentences, annotate the corresponding universal dependency relations on them.

Sentence	Relation
United canceled the flight	NSUBJ
United diverted the flight to Miami	DOBJ
We booked her the flight to Miami	IOBJ
We took the morning flight	NMOD
Book the cheapest flight	AMOD
JetBlue canceled 1000 flights	NUMMOD
The flight was canceled	DET
Book the flight through Houston	CASE
We flew Denver and Houston	CC, CONJ

VI. QUESTION (16 POINTS)

Given the following sentences, positive and negative categories; calculate the category for the given test sentences. Use Laplace smoothing to calculate probabilities.

Sentence	Category
Çok kötü film	-
Hep tekrar hep tekrar	-
Karakter gelişimi kötü	-
Süper güzel bir film	+
Yazın en iyi film bu	+
Çok güzel film	?
İyi karakter kötü film	?